Do Carbon Taxes Reduce CO_2 Emissions?

Cross-Country Panel Evidence, 1990–2024

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1. Introduction

Climate change and rising global CO_2 emissions represent one of the defining challenges of our time. Since the beginning of the Industrial Revolution, human activity has driven a dramatic increase in atmospheric carbon, contributing to ecological degradation, ocean acidification, and mounting threats to global food security, public health, and biodiversity. The primary market-based response to this crisis has been carbon pricing, implemented either as a direct carbon tax or through an emissions trading system (ETS). Despite steady growth in adoption, with carbon pricing now covering nearly 28% of global emissions compared to just 7% a decade ago (World Bank, 2025), emissions reached a record high of 37.8 billion tonnes in 2024 with no sign of peaking (IEA, 2025). Whether carbon pricing is actually driving meaningful reductions remains an open and consequential empirical question, and one this paper directly addresses.

History and Context

Carbon pricing first emerged in Scandinavia in the early 1990s, when Finland, Sweden, and Norway introduced carbon taxes in response to growing concerns over fossil fuel emissions. What began as a regional experiment gradually expanded into a broader global policy movement. By the 2000s and 2010s, emissions trading systems, most notably the European Union Emissions Trading System (EU ETS) launched in 2005, became the dominant model across much of the developed world. Two main approaches have since defined the carbon pricing landscape: carbon

taxes, which set a fixed price per ton of CO_2 emitted, and cap-and-trade systems, which impose a ceiling on total emissions and allow firms to buy and sell permits within that limit. Both mechanisms share the same economic foundation.

The theoretical case for carbon pricing rests on the concept of Pigouvian externalities, developed by economist Arthur Pigou. When firms emit CO_2, they impose costs on society through rising temperatures, extreme weather, and ecosystem damage that are not reflected in the market price of goods or energy. Left uncorrected, this leads to systematic overproduction of emissions relative to what would be socially optimal. A carbon price set equal to the marginal social damage of emissions forces firms to internalize these costs, creating direct incentives to reduce output, switch fuels, or invest in cleaner technology. In this sense, carbon pricing is not simply a revenue-generating tool; it is intended to correct a fundamental market failure.

From a policy standpoint, carbon pricing has become central to international climate commitments. The Paris Agreement, adopted by 195 parties at COP21 in 2015, established a framework of nationally determined contributions (NDCs) under which countries communicate their planned actions to reduce greenhouse gas emissions. More than half of these NDCs incorporate some form of carbon pricing, reinforcing its role as a leading policy instrument for achieving national climate targets (UNFCCC, 2015).

The empirical evidence on the effectiveness of carbon pricing remains promising but mixed. Best et al. (2020), in one of the most comprehensive cross-country studies to date, found that countries with carbon pricing policies experience CO_2 emissions growth rates roughly 2 percentage points lower than countries without such policies, after controlling for income, energy structure, and other environmental policies. More recently, Lamb et al. (2024) conducted a meta-analysis of 21 carbon pricing schemes and found statistically significant emissions reductions ranging from 5 to 21%, although the magnitude of these effects varies substantially depending on price levels, sectoral coverage, and national context. A recurring theme in the literature is that existing carbon prices remain too low to achieve global climate targets. According to the World Bank (2024), less than 1% of global greenhouse gas emissions are covered by a carbon price at or above the level estimated to limit warming to well below 2°C. This paper contributes to the existing literature by using updated panel data spanning 1990 to 2024 and a two-way fixed effects framework across 215 countries, aiming to provide updated

cross-country evidence on whether higher carbon prices are associated with lower CO_2 emissions per capita.

2. Research Question

This paper examines whether a higher carbon price leads to lower CO_2 emissions per capita at the country level. Based on Pigouvian theory, the expectation is that a higher carbon price raises the cost of emitting, incentivizing firms and households to reduce emissions through fuel switching, energy efficiency, and investment in cleaner technology, producing a negative relationship between carbon price and per capita emissions. This question is tested empirically using a two-way fixed effects panel regression across 215 countries from 1990 to 2024, controlling for GDP per capita and energy intensity. The remainder of the paper is structured as follows: Section 3 describes the data, Section 4 presents the theoretical framework, Section 5 outlines the empirical methodology, Section 6 reports the main results, and Section 7 concludes.

3. Description of Data

Overview and Sources

This paper draws on two publicly available datasets that together provide the coverage and variable depth needed to study the relationship between carbon pricing and CO_2 emissions at the national level over a 35-year horizon.

The primary source for emissions and macroeconomic data is the Our World in Data (OWID) CO_2 and Energy dataset, which compiles information from the Global Carbon Project, the International Energy Agency, and the World Bank. The dataset covers 218 countries from 1750 to 2024 and provides annual country-level observations on CO_2 emissions per capita (measured in metric tonnes), total GDP, population, and energy intensity (measured as primary energy consumption per unit of GDP). For this study, the dataset is restricted to 1990 onward to align with the start of the carbon pricing era, and regional aggregates such as "World," "High-income countries," and continental groupings are dropped, leaving a sample of 215

individual countries as they would create issues of double counting and bias. GDP per capita is constructed by dividing total GDP by population.

Carbon pricing data is sourced from the World Carbon Pricing Database (WCPD), published by Resources for the Future (RFF) and released through August 2025. This database records compliance carbon prices in USD per ton of CO_2 equivalent for 80 national and subnational instruments from 1990 to 2025. For this analysis, subnational jurisdictions including Canadian provinces such as Alberta and British Columbia, U.S. states such as California, and Chinese cities and provinces are excluded in order to maintain the analysis at the country level. For instances where a country operates multiple systems simultaneously, such as both a carbon tax and an ETS, the maximum observed price in a given year is kept, yielding a single carbon price observation per country per year. Countries with no active carbon pricing instrument in a given year are assigned a price of zero, reflecting the absence of any explicit carbon cost and are not considered in the final regression.

The two datasets are merged on country name and year, producing an unbalanced panel(not all units are observed). The final dataset contains 5,358 country-year observations across 215 countries spanning 1990 to 2024. Of these, 37 countries have at least one year with a carbon price above zero.

Main Variables

The dependent variable is CO_2 emissions per capita, measured in metric tonnes. This variable captures country-level emissions on a comparable per-person basis, making it suitable for cross-country analysis where countries differ substantially in population size. The key independent variable is the carbon price, measured in current USD per ton of CO_2 equivalent. This variable directly captures the stringency of the carbon pricing policy in place and is the central treatment of interest. Two control variables are included in the regression. GDP per capita, in constant USD, controls for the strong positive relationship between income and energy consumption, which would otherwise confound the estimated effect of carbon pricing. Energy intensity, measured as kilowatt-hours of primary energy consumption per dollar of GDP, controls cross-country differences in the structure of the economy; more energy-intensive economies tend to emit more regardless of whether a carbon price is in place.

Both the outcome variable and the key independent variable are log-transformed prior to estimation. Because many country-years have a carbon price of zero, the transformation applied is log(carbon price + 1), which preserves the zero observations while allowing a log-log elasticity interpretation of the coefficient. Under this specification, the coefficient on log carbon price can be read as the percentage change in CO_2 per capita associated with a 1% increase in the carbon price. This serves the purpose to help compress values to prevent outlier domination, help better capture diminishing proportional effects and will help to reduce variance ameliorating heteroskedasticity issues.

Data Type and Unit of Observation

The dataset is a cross-country panel, with the unit of observation being a country in a given year. This structure allows the use of country fixed effects, which absorb all stable country-level characteristics such as geography, historical energy mix, institutional quality, and culture, and year fixed effects, which control for common global shocks such as recessions or the COVID-19 pandemic. Together, these fixed effects isolate the within-country variation in carbon prices over time as the identifying source of variation for the estimated effect on emissions.

Descriptive Statistics

Table 1 in Appendix 1 presents summary statistics for the four main variables across the regression sample of 5,358 country-year observations. The average CO_2 emissions per capita is 4.96 metric tonnes, with substantial variation ranging from near zero in low-income countries to nearly 360 metric tonnes in small-high income outlier countries. The average GDP per capita is approximately $14,370, again with wide cross-country dispersion. Energy intensity averages 1.54 kWh per dollar of GDP across the sample, showing significant energy efficiency discrepancies across countries. Figure 1 in Appendix 1 presents a scatter plot of the logged carbon price against logged CO_2 emissions per capita across all country-year observations, providing a visual illustration of the unconditional relationship between carbon pricing and emissions. Figure 2 plots average CO_2 emissions per capita over time separately for countries with and without a carbon price, illustrating differences in emissions trajectories that motivate the regression analysis.

4. Theoretical Framework

The theoretical foundation of this paper comes from the idea of negative externalities, introduced by economist Arthur Pigou. A negative externality occurs when the production or consumption of a good creates costs for others that are not included in the market price. Carbon emissions are a clear example because firms that burn fossil fuels contribute to climate change, creating social costs such as environmental damage, public health concerns, and extreme weather not being directly paid by these firms. As a result, these firms produce more emissions than is socially optimal or more commonly the marginal social cost does not equal marginal social benefit.

Pigou argued that governments can correct this market failure with a tax equal to the social cost of the externality. In the case of carbon emissions, this means placing a price on each ton of CO_2 emitted. Economic theory suggests that higher carbon prices reduce emissions because carbon intensive activities become more expensive, encouraging firms and consumers to switch to cleaner alternatives and invest in low carbon technologies.

Based on this framework, this paper expects a negative relationship between carbon prices and CO_2 emissions per capita. Higher carbon prices should lead to lower emissions. To isolate this effect, the analysis controls for GDP per capita, since wealthier countries tend to consume more energy, and for energy intensity, since more energy intensive economies naturally produce higher emissions regardless of carbon pricing systems in place.

5. Data Analysis

This paper estimates the effect of carbon pricing on CO_2 emissions per capita using a two-way fixed effects panel regression. The core model is:

$$log(CO_2\ per\ capita)_{i} =$$

$$\beta\, log(carbon\ price\ +\ 1)_{it} + \gamma\, log(GDP\ per\ capita)_{it} + \delta\, log(energy\ intensity)_{it} + \alpha_i + \lambda_t +$$

The variable of concern is β, which measures the relationship between carbon price and emissions within a country over time. Country fixed effects (αi) control for stable national characteristics like geography, culture, and historical energy infrastructure which can be seen in the Stata code. Fixed year effects (λt) control for global events that affect all countries simultaneously, such as the 2008 financial crisis or the COVID-19 pandemic. All models are estimated in Stata using the xtreg command with country fixed effects and robust standard errors.

Both the dependent variable and the carbon price are log-transformed. Because many countries have a carbon price of zero, the transformation log(carbon price + 1) is applied, which handles zero values while still allowing the coefficient to be interpreted as an elasticity as explained earlier in the paper. This means β represents the percentage change in CO_2 per capita for a 1% increase in carbon price.

Three models are estimated in sequence. Model 1 includes only the carbon price and country fixed effects. Model 2 adds GDP per capita as a control. Model 3 is the full specification combining the previous models adding energy intensity and year fixed effects. This smooth progression allows us to see how the carbon price coefficient changes as controls are added and whether the earlier estimates were biased by omitted variables. A robustness check (Model 4) repeats the full specification using only the 37 countries that ever implemented a carbon price.

6. Main Results

The results I will be focused on are the results of the regression taken from Table 2. As expected the carbon price coefficient is negative and all of the results are statistically significant at the 1% level. In the most extensive model (Model 3), the coefficient is extremely close to Model 1. The coefficient is -0.064 meaning a 10% increase in the price of carbon is associated with a 0.64% reduction in CO2 emissions per capita, controlling for income, country characteristics, global trends, and energy intensity.

As we see the spike in the negative coefficient going from -0.063 to -0.109, we can explain this spike through the omission of energy intensity allowing for GDP to absorb some of the variation. Moving away from the coefficient of concern, the R^2 warrants attention . We can see it progressively increasing from 0.01 to 0.598 explaining more and more of the variation from 1% to 60% of the variation.

Furthermore, the control variables are predictable and behave perfectly. Taking from model 3, the coefficient is 0.810, which as expected wealthier countries would emit more. Additionally, log energy intensity is 0.672, meaning energy intensive produce higher emissions.

The last table to mention is table 3 representing the robustness check. The most interesting finding would be the coefficient dropping to -0.11 and becoming not statistically significant. This does not invalidate my previous findings as the sample size drops to 1,056 meaning that the sample size might not be large enough to detect a significant price level variation. Lastly, it is expected that log GDP and log energy intensity would increase as these countries would be expected to be more affluent as adopting a carbon tax is typically for those well into industrialization.

7. Conclusion

In conclusion, this paper examined whether higher carbon prices are associated with lower CO_2 emissions per capita using a panel of 215 countries from 1990 to 2024. My results supported the Pigouvian theory hypothesis with a 10% increase in carbon price being associated with a 0.64% reduction in CO_2 emissions per capita. This finding is statistically significant at the 1% level and includes controls of GDP per capita, energy intensity, and country/year fixed effects. However, the most interesting finding is found by restricting to 37 countries which are all countries that currently have carbon pricing implemented. The effect disappears entirely and suggests that the effect is driven by the sheer difference of countries that do or do not have carbon pricing. Additionally, the robustness check allows for a more nuanced position that aligns with the World Bank's findings that adopting a carbon price is more important than the value as

the values are not significant enough to gauge whether or not the price level lowers emissions as seen by the coefficient of concern being -0.011 and statistically insignificant.

These findings are clear-cut regarding policy implementation. We should expand carbon pricing to more countries rather than focusing on increasing prices in countries with pre-existing carbon pricing. This does not come without caveats, as countries with and without carbon pricing differ greatly, we have to account for other uncontrolled variables such as pre-existing climate policy and wealthier environmental institutions compared to any at all. Nevertheless, this paper provides evidence for the broader case of global carbon pricing as a core instrument of climate change prevention and highlights the necessity for continued policy adoption in countries that have yet to implement a price on carbon emissions.

References

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Table 1:

Descriptive Statistics					
Variable	Mean	Std. Dev	Min	Max	N
CO2 per Capita	4.96	8.34	0.02	364.79	5,358
Carbon Price	2.08	12.79	0.00	168.83	5,358

(USD per tCO_2)					
GDP per Capita (USD)	14,369.95	16,416.17	361.19	163,531.41	5,358
Energy Intensity	1.54	1.47	0.08	23.02	5,358

**is measured as kilowatt-hours of primary energy consumption per dollar of GDP*

**Carbon Price measured in USD per metric ton of CO2 Emissions*

Table 2: Regression Output

Variables	Model 1	Model 2	Model 3
Key Variable			
log(carbon price + 1)	-0.063***	-0.109***	-0.064***
SE	(0.012)	(0.012)	(0.010)
Controls			
log GDP per capita	—	0.471***	0.810***
SE		(0.068)	(0.058)
log energy intensity	—	—	0.672***
SE			(0.058)
Fixed Effects			
Country FE	Y	Y	Y
Year FE	N	N	Y
Fit Statistics			
N	5,358	5,358	5,358
R^2	0.010	0.259	0.598

Adjusted R^2	0.010	0.258	0.596

** Dependent variable is log CO_2 per capita. All models use country fixed effects estimated with Stata xtreg, fe and robust standard errors in parentheses. Model 3 also includes year fixed effects. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$*

Table 3: Robustness Regression Check

Variables	Full Sample	Priced Countries Only
Key Variables		
log(carbon price + 1)	−0.064***	−0.011
SE	(0.010)	(0.009)
Controls		
log GDP per capita	0.810***	0.950***
SE	(0.058)	(0.123)
log energy intensity	0.672***	0.782***
SE	(0.058)	(0.100)
Fixed Effects		
Country FE	Y	Y
Year FE	Y	Y
Fit Statistics		
N	5,358	1,056
R^2	0.598	0.721

Figure 1 : Carbon Pricing and Co2 Emissions per Capita (log)

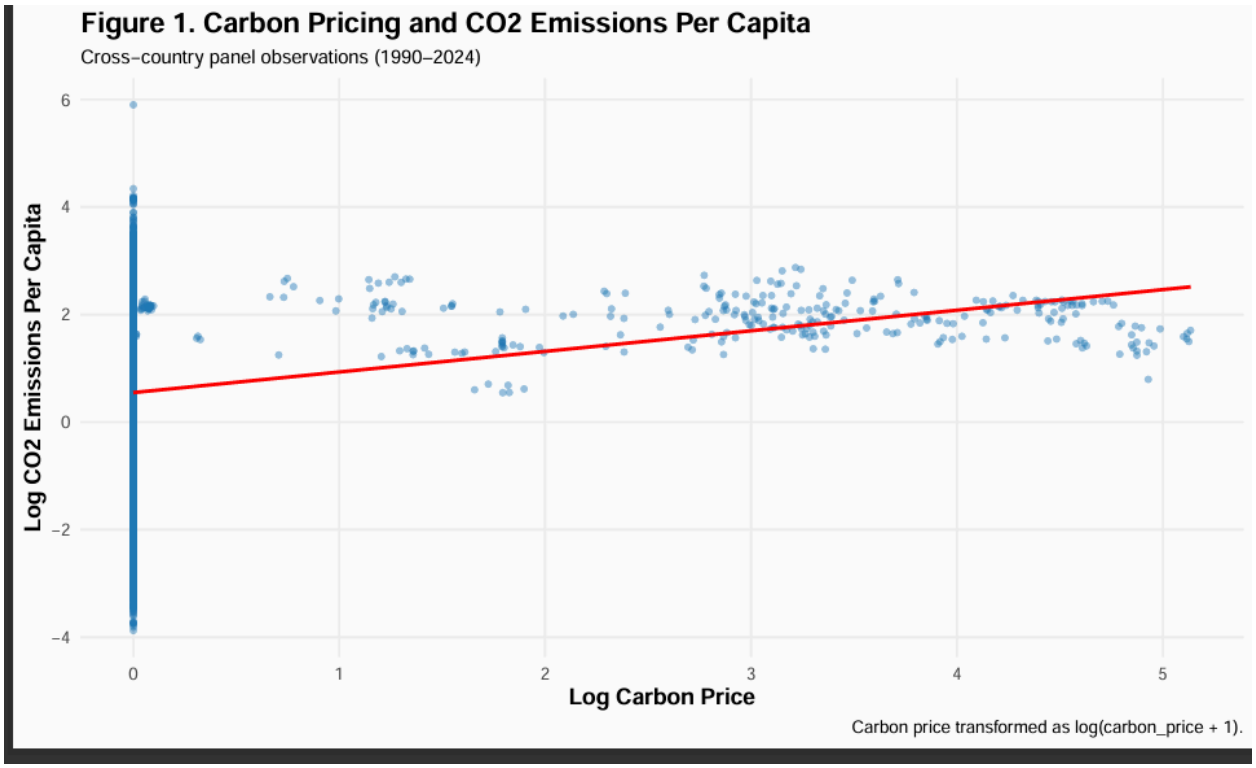

Figure 2: Trends by Carbon Pricing Status

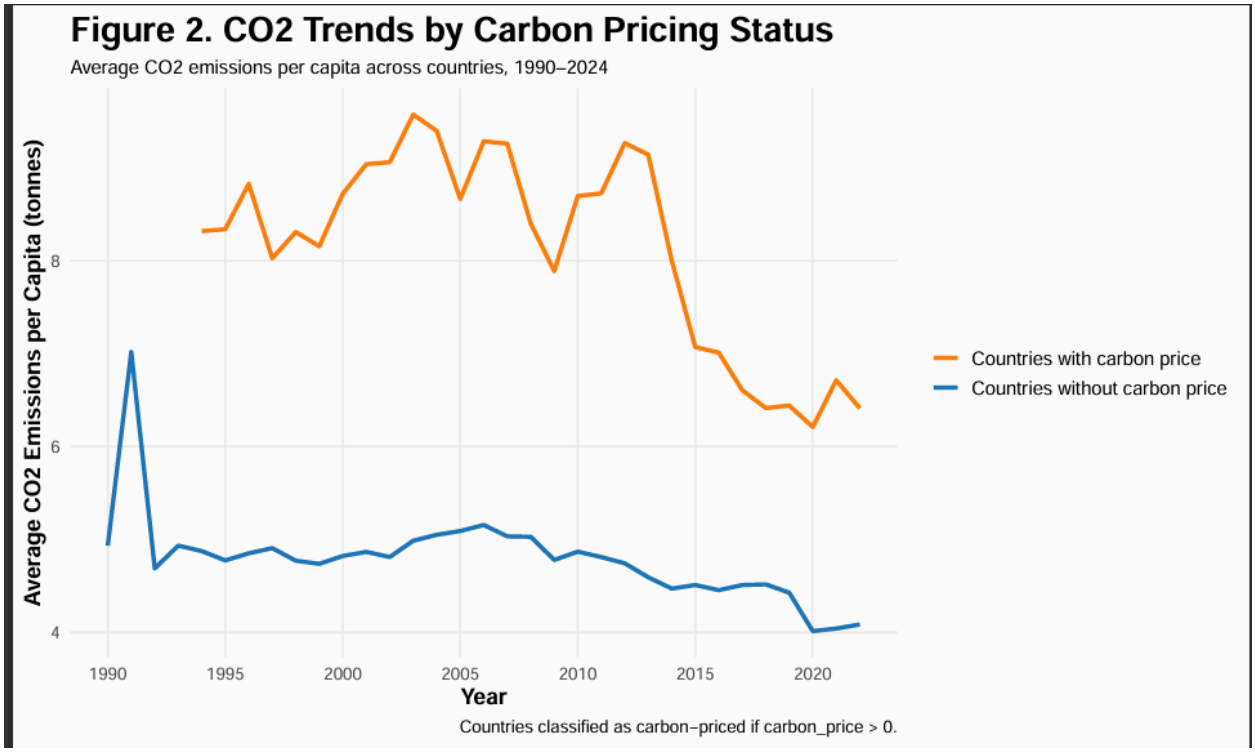
Figure 2. CO2 Trends by Carbon Pricing Status
Average CO2 emissions per capita across countries, 1990–2024
Average CO2 Emissions per Capita (tonnes)
8
6
4
1990
1995
2000
2005
2010
2015
2020
Year
Countries classified as carbon–priced if carbon_price > 0.
Countries with carbon price
Countries without carbon price